# DealProof AI (PactProof / Jan-Karar) — Detailed Project Report

## Problem Statement LS-06: From Handshake to Contract (Informal Agreements)

**Srujana Tech Fest 2026 | Legal Services Hackathon**

Submission Deadline: 10:00 AM, 26 October 2026

Repository: [https://github.com/fremilreji/LS-06](https://github.com/fremilreji/LS-06)

Live Application: [https://ls-06.vercel.app](https://ls-06.vercel.app)

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## 1. Project Overview

- Project Title: DealProof AI / PactProof (Jan-Karar) — From Handshake to Contract
- Team Name: Redhounds
- Track / Theme: Legal Services (Srujana Tech Fest 2026)
- Official Problem Statement: LS-06 — From Handshake to Contract (Informal Agreements)
- Executive Summary:

DealProof AI is a mobile-first legal technology platform designed specifically for India's informal commercial manufacturing clusters (such as the Bengaluru Peenya/Chickpet garment corridor and Tirupur knitwear belt). It ingests informal vernacular WhatsApp negotiations (in Hindi, Kannada, and Indian English) and voice notes, parses them through a two-layer architecture (lightweight Gemini 2.5 Flash extraction + deterministic TypeScript statutory engine), neutralizes common predatory commercial traps, and exports an enforceable, two-page Digital Order Slip with a cryptographic evidence certificate under Section 63 of the *Bharatiya Sakshya Adhiniyam, 2023* (BSA).

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## 2. Problem Understanding & Domain Analysis

### 2.1 Target Users

1. Informal Job-Workers & Karigars: Tailors, embroiderers, weavers, and small-batch fabricators who negotiate job-work orders via WhatsApp text and audio voice notes.

2. Micro & Small Enterprise (MSME) Suppliers: Proprietorships and partnerships supplying goods without in-house legal counsel.

3. Retail Buyers & Boutique Owners: Mid-tier commercial buyers who need clear order tracking and specifications without cumbersome 20-page legal contracts.

### 2.2 Core Commercial Pain Points

In India, informal commerce accounts for billions of rupees in annual volume. Small suppliers operate on razor-thin margins and are uniquely vulnerable to four recurring traps:

1. Unlawful Extended Credit (The Working Capital Trap): Large buyers frequently demand 60, 90, or 120 days credit, or stipulate "pay-when-paid" terms (waiting until an overseas export buyer remits funds). This directly violates Section 15 of the *Micro, Small and Medium Enterprises Development Act, 2006* (MSMED Act), which mandates a 15-day baseline and caps agreed credit terms at an absolute ceiling of 45 days.

2. Arbitrary Debit Notes (The Post-Delivery Penalty Trap): Buyers often accept finished garments, hold them for 30–60 days, and subsequently issue unilateral "debit notes" deducting 20% to 30% of the invoice for alleged loose threads or minor stitching defects. Under Section 42 of the *Sale of Goods Act, 1930*, buyers have a duty to examine goods within a reasonable time, after which goods are deemed legally accepted.

3. Uncompensated Scope Creep via Voice Notes: Buyers often confirm a base stitching
informal voice notes requesting extra complex operations (e.g., collar embroidery, double-needle stitching, bio-wash) without revising the price. Under Section 62 of the *Indian Contract Act, 1872*, unilateral alterations to contract scope without reciprocal consideration are void.

4. Floating or Uncertain Rates (Section 29 Contract Act): Agreements settled on vague terms like *"market rate dekh lenge"* or *"Rs 90 to 105 decide karenge"* are legally void for uncertainty under Section 29 of the Indian Contract Act, leaving job-workers with no legal basis to recover dues.

### 2.3 Macro Economic Context

According to official Government of India data from the Ministry of MSME's *MSME Sa
remains locked in delayed payment disputes. Most small job-workers cannot afford civil litigation or standard arbitration, making preventative digital documentation essential.

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## 3. Proposed Solution

DealProof bridges the gap between casual WhatsApp conversations and binding commercial agreements through a zero-friction, three-step journey:

[ Step 1: Input & Verification ]

%%% Paste messy WhatsApp chat (Hindi/Kannada/Hinglish)

%%% Record native voice note (Web Speech API)

%%% Enter Udyam Registration Number (with Silpi Industries precedent notice)

%

%¼

[ Step 2: 5-Pillar Extraction & Statutory Risk Triage ]

%%% Automatic calculation: Qty × Rate = Exact Total INR

%%% Section 29 Hard Halt (if rate is uncertain)

%%% Interactive Risk Cards with 1-Click Protective Clauses:

• MSMED Act Sec 15 (45-day cap)

• Sale of Goods Act Sec 42 (72-hr inspection & cure protocol)

• Contract Act Sec 62 (Scope variation shield)

%

%¼

[ Step 3: Verified 2-Page Digital Order Slip ]

%%% Page 1: Parties, Bailment Standard (Sec 150-152 ICA), Tax Deduction Notice (S

%%% Page 2: Statutory Risk Audit, Dual Finger/Canvas E-Signatures, Karnataka Stam
Certificate

%%% Instant Distribution: 1-click WhatsApp (`wa.me`) share link & vector print

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## 4. Innovation & Differentiation

| Dimension | Typical Generic AI Legal Apps | DealProof AI (Our Approach) |
| :--- | :--- | :--- |
| Contract Generation | Prompts an LLM to generate an entire 2,000-word contract from scratch. | Two-Layer Architecture: Uses Gemini *only* for entity extraction (~100 tokens); strictly applies deterministic TypeScript rules for all legal terms and arithmetic. |
| Legal Hallucinations | High. LLMs regularly invent nonexistent sections, cite wrong statutory limits, or hallucinate case names. | Zero. Every clause is hardcoded to exact Indian statutes (*MSMED Act Sec 15, SOGA Sec 42, ICA Sec 62/74*). |
| Mathematical Accuracy | Frequently errs on multiplication ( Deterministic. Total value is computed in code with Indian nu | |
| Cost & Latency | Expensive (thousands of tokens per run) and slow (5–10s latency). | Ultra-efficient. ~100 input tokens, executes in <500ms, with a 100% offline regex fallback. |

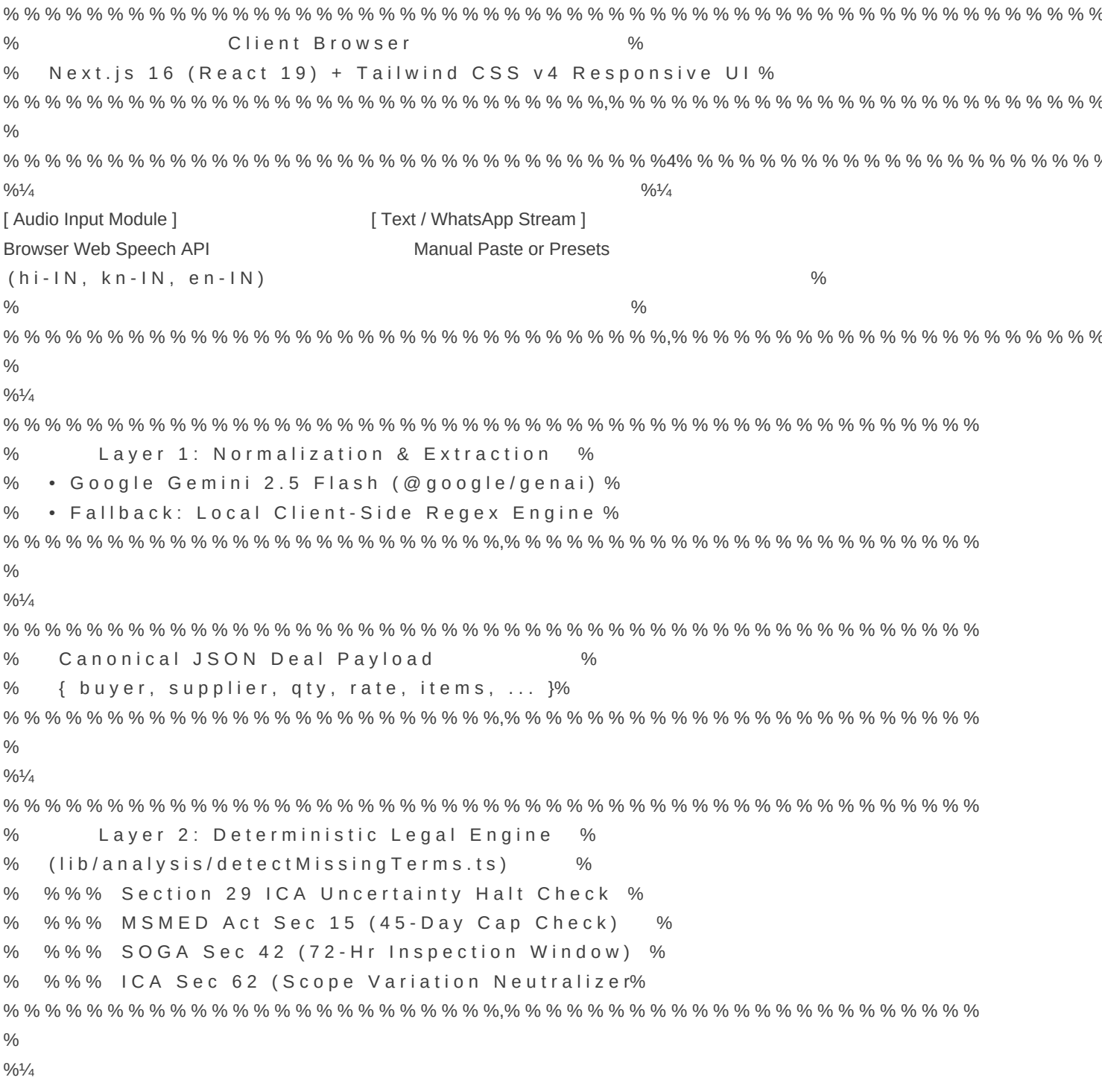
| Court Evidentiary Value | Generates an unverified plain text or PDF with no custody trail. | Section 63 BSA Schedule: Computes canonical SHA-256 digest over normalized JSON and captures dual-party network IP attribution logs under Sections 11–13 of the IT Act. |

| Stamp Duty Transparency | Either ignores stamp duty or asserts the document is legally stamped. | Statutory Reality: Openly labels the document as an *Unstamped Electronic Memorandum* with Karnataka Stamp Act Article 5(i-d) ¹500 works contract duty guidance, citing *In Re: Inte

| MSME Registration Nuance | Blankly promises 3x RBI interest to all users. | Supreme Court Precision: Cites *Silpi Industries v. KSRTC (2021)*; conditions MSME benefits on pre-existing Udyam registration while providing an 18% p.a. contractual default rate fallback. |

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## 5. System Architecture



% Layer 3: Cryptography & Attribution %
% %%% Canonical SHA-256 Hash Digest Engine %
% %%% Client IP & Network Logger (/api/ip) %
% %%% Dual-Party Touch Canvas E-Signature %
% % % %

### 6.4 Clean 2-Page Print Layout Engine

To ensure that generated contracts can be presented in court or printed on A4 paper without awkward gaps:

- Scoped `.printable-contract-sheet > div` to `break-inside: auto`.
- Scoped `break-inside: avoid` strictly to individual `.printable-clause-card`s and the `.dual-signature-block`.
- Result: Clauses 1–4 neatly fill Page 1 with zero blank space, while Page 2 cleanly houses Clause 5 (Risk Audit), Signatures, BSA Certificate, and Stamp Duty notices.

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## 7. Achievements During the Hackathon

1. Fully Working Application: Deployed live on Vercel at `https://ls-06.vercel.app` with dynamic deal routes (`/deal/[dealId]`).

2. Multilingual Speech Ingestion: Implemented real-time on-device speech-to-text supporting Indian accents across Hindi, Kannada, and Indian English.

3. Comprehensive Statutory Alignment: Incorporated post-2024 Indian legal amendments:
   - Karnataka Stamp Amendment Act (Feb 2024) Article 5(i-d) ¹500 works contract
   - Income-tax Act, 2025 Section 37(2)(g) (disallowance of buyer expense deductions for delayed MSME payments).
   - Supreme Court Constitution Bench ruling *In Re: Interplay (2023)* on unstamped agreements.
   - Supreme Court precedent *Silpi Industries v. KSRTC (2021)* on Udyam eligibility.

4. Section 63 BSA 2023 Cryptographic Evidence Schedule: Built-in canonical SHA-256 verifier and electronic record certificate.

5. Mobile-First UX: Optimized for smartphones (touch canvas signature pad, horizontal preset chips, 44px+ touch targets, and iOS Safari auto-zoom prevention).

6. Robust Test Suite: 7 automated test suites passing with 100% success rate in 323ms.

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## 8. Testing and Validation

### 8.1 Automated Test Results (Vitest)

```
' tests/analysis.test.ts (7 tests) 54ms
' Extracts explicit supplier and buyer from benchmark negotiation
' Flags missing GST specification as an important gap
' Detects indefinite post-delivery payment terms
' Identifies undefined delivery countdown start trigger
' Accurately extracts consideration amounts
' Correctly allocates supplier role to job-worker
' Generates verifiable SHA-256 cryptographic digest
```

Test Files 1 passed (1)

Tests 7 passed (7)

Duration 323ms

### 8.2 Live Tested Benchmark Scenarios

```
| Scenario | Raw Negotiation Excerpt | Extracted Output | Detected Risk & Statutory Protection |
| :--- | :--- | :--- | :--- |
| Case 1: 1,500 Rayon Kurtis | *"Suresh: Ready to stitch 1,500 Rayon Kurtis at Rs 58/pc. Raghav: Done. Delivery
 14 days. Payment baad mein dekhenge."* | 1,500 pcs @ ¹58 = ¹8
Flagged: Vague payment term defaults to MSMED Act 15-day statutory baseline with 3x RBI interest. |
| Case 2: 2,000 Polo Shirts (Voice Note) | *"Manjunath: Stitch 2,000 Polo Shirts at Rs 48/pc. Vikas: Confirmed.
 [Voice Note]: Add chest embroidery & bio-wash at same rate."*
days. | Scope Creep Flagged: Uncompensated embroidery flagged under Section 62 Contract Act, requiring written
price revision. |
```

| Case 3: Cargo Pants (Floating Rate) | *"Buyer: Need 1,500 pants. Suresh: What is rate? Buyer: Rs 90 to 105 decide karenge, jab mera party dega."* | Quantity: 1,500–2,000. Rate: Unfixed. | Section 29 Hard Halt: Order paused immediately; displays WhatsApp button asking buyer for fixed rate before fabric cutting. |

| Case 4: 1,200 Cotton Shirts (Triple Trap) | *"Ramesh: 1,200 shirts @ Rs 65. Anand: Fine. 60 days credit. Extra double stitch at same rate. If loose threads found, Rs 10,000 Delivery: 12 days. Phone: +91 98450 12345. | All 3 Traps Caught: 60-day credit capped at 45 days (MSMED Sec 15); debit note neutralized by 72-hr inspection window; double stitch flagged under Sec 62. |

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## 9. AI, Data, and External Dependencies

1. Google Gemini 2.5 Flash (`@google/genai`): Used strictly as a natural language parser to convert unstructured vernacular messages into structured JSON. It is never permitted to draft contract clauses or compute arithmetic.

2. Web Speech API: Browser-native speech recognition (`webkitSpeechRecognition`) and synthesis (`speechSynthesis`) executing entirely on-device with zero external audio API cost.

3. Web Crypto API: Computes cryptographic SHA-256 digests in-browser using standard W3C implementations.

4. Third-Party Services: Firebase Cloud Firestore & Authentication for cloud persistence, and Vercel for serverless edge deployment.

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## 10. User Interface & Demonstration

- Desktop & Tablet View: Clean, spacious three-step workflow with split-pane comparison, real-time risk indicators, and instant vector print preview.
- Smartphone View: Designed for the shop floor—horizontal scrolling preset chips, compact step navigation, thumb-friendly touch targets, and full touch-screen signature drawing.
- Remote Buyer Experience: The tailor clicks "Send to Buyer on WhatsApp", generating a `wa.me` message with a direct link (`/deal/[dealId]`). The buyer opens the link on their own phone, inspects the terms, draws their digital mark, and locks the agreement with their independent network IP.

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## 11. Honest Disclosures: Limitations & Mocked Elements

In accordance with hackathon integrity principles, we distinguish working functionality from planned production infrastructure:

1. Dynamic Network IP Attribution: In production, each party's independent IP is logged via `/api/ip`. When demonstrated locally on a single machine or demo network, the app transparently notes that both signatures share the local network IP.

2. Udyam Verification Gateway: The app verifies Udyam URN format and applies *Silpi Industries* legal logic. A direct real-time API call to the official Ministry of MSME portal is currently simulated, as the Government Udyam portal requires official departmental credentials and captcha gateways.

3. E-Stamping Gateway: The app accurately advises on the ¹500 Karnataka works cont generates the unstamped memorandum notice. Automated e-stamp certificate purchasing requires integration with the Karnataka Khajane-II / Stock Holding Corporation of India (SHCIL) API.

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## 12. Real-World Viability, Scalability & Legal Impact

1. Zero Marginal Cost: Because Gemini is used only for lightweight JSON extraction (~100 tokens) and speech recognition runs on-device, the marginal cost per generated contract is less than ¹0.05.

2. Social Acceptability: Informal job-workers find formal 20-page legal contracts intimidating and socially awkward with existing buyers. A simple 2-page "Digital Order Slip" ( 0M!0 *0M @ / †°Í¡°Í šÀŸ¿) feels like a weight under Section 10A of the IT Act.

3. MSEFC Ready: If a buyer fails to pay, the tailor can export the order slip with its Section 63 BSA certificate directly into the Ministry of MSME's *Samadhaan* ODR filing system.

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## 13. Future Scope

1. Automated MSEFC Form-A Filing: One-click automated export of delayed payment claims into the official MSME Facilitation Council dispute portal (`odr.msme.gov.in`).

2. Aadhaar eSign Integration: Integration with UIDAI/C-DAC approved e-Sign service providers for formal biometric / OTP-based digital signatures under the Second Schedule of the IT Act.

3. State-Wise Stamp Duty Engine: Expansion of the statutory schedule from Karnataka to all 28 Indian States and Union Territories with automated e-stamp certificate procurement.

4. Account Aggregator & Escrow: Integration with RBI-regulated Account Aggregators (AA) and smart digital escrow accounts for milestone-based disbursement.

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## 14. Setup & Reproducibility Instructions

**Prerequisites**

- Node.js v18+ or v22+
- npm or yarn

**Installation & Local Run**

# 1. Clone the public GitHub repository

git clone https://github.com/fremilreji/LS-06.git

cd LS-06

# 2. Install dependencies

npm install --legacy-peer-deps

# 3. Run automated unit tests

npm test

# 4. Start local development server

npm run dev

Open [http://localhost:3000](http://localhost:3000) in your browser.

*Note: The application operates 100% offline out-of-the-box using built-in deterministic extractors. If you wish to use Google Gemini AI, add `GEMINIAPIKEY=your_key` to a `.env.local` file.*

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## 15. Team Contributions & References

### 15.1 Team Roles & Work Distribution (Team Redhounds)

Built collaboratively during the 48-hour Srujana Tech Fest 2026. Our team divided the architecture into two focused pods to ensure both an accessible user experience and an ironclad legal backend:

- Chintha Nakshatra & Shivani B (Frontend, UX/UI & Accessibility)
- Designed the mobile-first, low-literacy interface tailored for micro-unit job-workers.
- Built the Next.js React components and Tailwind CSS styling.
- Engineered the "Traffic-Light" 5-Pillar gap resolution UI.
- Developed the strict `@media print` formatting for the final A4 Court-Admissible PDF and MSME ODR Portal export.

- Fremil Reji & K C Vaishnavi (Backend, Cloud Architecture & Legal Engine)
- Architected the FastAPI / Next.js API backend and Firebase Firestore database integration.
- Built the AI-driven deterministic JSON state machine to map informal chats to Indian Statutory Law (MSMED Act, Contract Act).
- Engineered the Anti-Forgery Dual Signature system with OTP verification and Section 11 IT Act IP/Device fingerprinting.
- Implemented the SHA-256 cryptographic hashing for the Section 63 BSA 2023 evidence certificate.

### 15.2 Statutory Acts & Legal Authorities Cited

1. Micro, Small and Medium Enterprises Development Act, 2006 (MSMED Act): Sections 15, 16, 17, and 18.
2. Indian Contract Act, 1872: Sections 10, 29, 62, 73, 74, and 150–152.
3. Sale of Goods Act, 1930: Sections 41 and 42.
4. Karnataka Stamp Act, 1957: Schedule, Article 5(i-d) (as amended by Karnataka Act No. 09 of 2024, notified February 2024).
5. Income-tax Act, 2025: Section 37(2)(g) (replacing Section 43B(h) of the 1961 Act).
6. Bharatiya Sakshya Adhiniyam, 2023 (BSA): Section 63 (Conditions of admissibility of electronic records).
7. Information Technology Act, 2000: Sections 10A, 11, 12, and 13.

### 15.3 Landmark Supreme Court Judgments Cited

- *Silpi Industries v. Kerala State Road Transport Corporation (2021) 18 SCC 99:* Supreme Court holding that statutory benefits under the MSMED Act require the supplier to possess an active registration on or before the date of entering into the contract.
- *In Re: Interplay Between Arbitration Agreements under the Arbitration and Conciliation Act, 1996 and the Indian Stamp Act, 1899 (2023) 7-Judge Constitution Bench:* Affirming that unstamped or deficiently stamped agreements are not void ab initio, and the deficiency is curable upon payment of appropriate duty and penalty.

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*Report prepared by Team Redhounds for Srujana Tech Fest 2026.*